

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 01

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Two similar solids have surface areas in the ratio 4 : 9. Find the ratio of their volumes.

A 12 : 27

B 16 : 81

C 2 : 3

D 8 : 27

E 64 : 729

F 4 : 9

2 Express the recurring decimal $0.\dot{2}\dot{7}$ as a fraction in its lowest terms.

- A** $\frac{9}{33}$
B $\frac{18}{3}$
C $\frac{11}{27}$
D $\frac{99}{3}$
E $\frac{10}{27}$
F $\frac{100}{100}$

3 Given that $\log_{10} 2 = p$ and $\log_{10} 3 = q$, express $\log_{10} 0.75$ in terms of p and q .

A $3q - 4p$

B $\frac{q}{2p}$

C $2p - q$

D $q + 2p$

E $q - p$

F $q - 2p$

- 4 How many real solutions does the equation $4^x + 2^{x+1} - 8 = 0$ have?
- A 1
 - B 2
 - C 0
 - D 3
 - E infinitely many

- 5 AB is a diameter of a circle and C is a point on the circle with $AC = 6$ and $BC = 8$. Find the radius of the circle.

A $\sqrt{50}$

B 4

C 10

D 7

E 5

F 14

6 The coefficient of x^2 in the expansion of $(1 + kx)^6$ is 60. Given that $k > 0$, find k .

A 2

B 4

C 3

D $\sqrt{10}$

E $\frac{1}{2}$

F 10

- 7 The sequence $u_n = n^2 - 10n + 30$ is defined for positive integers n . What is the smallest value taken by any term of the sequence?
- A** 21
- B** 5
- C** 6
- D** -5
- E** 0
- F** 30

8 Solve $9^x - 4 \times 3^x + 3 = 0$.

A $x = 1$ only

B $x = 0$ or $x = 1$

C $x = -1$ or $x = 0$

D $x = 0$ only

E $x = \log_3 4$

F $x = 1$ or $x = 3$

9 $f(x) = ax + b$ satisfies $\int_0^1 f(x) \, dx = 2$ and $\int_0^2 f(x) \, dx = 6$. Find $f(1)$.

A 4

B 0

C 5

D 1

E 3

F 2

10 For how many integers n is $n^2 - 10n + 16 < 0$?

A 5

B 3

C 4

D 8

E 7

F 6

11 Find the value of $\tan 30^\circ \tan 60^\circ + \sin^2 45^\circ$.

A $\frac{\sqrt{2}}{2}$

B 1

C $\frac{3}{2}$

D 2

E $1 + \sqrt{3}$

F $\frac{1}{2}$

12 Given that $v^2 = u^2 + 2as$, find s when $v = 13$, $u = 5$ and $a = 4$.

A 18

B 21

C 36

D 12

E 9

F 24

13 In triangle ABC , $AB = 5$, $AC = 8$ and angle $BAC = 60^\circ$. Find BC .

A $\sqrt{129}$

B $\sqrt{49 + 40\sqrt{3}}$

C $\sqrt{89}$

D $\sqrt{69}$

E 9

F 7

14 Find the complete set of values of x for which $2x^2 - 5x - 3 \geq 0$.

A $x \leq -\frac{1}{2}$ or $x \geq 3$

B $x \leq -\frac{1}{2}$

C $x \geq 3$

D $x \leq -3$ or $x \geq \frac{1}{2}$

E $-3 \leq x \leq \frac{1}{2}$

F $-\frac{1}{2} \leq x \leq 3$

- 15 A geometric series has first term a and common ratio r with $|r| < 1$. Its sum to infinity is 8 and the sum to infinity of the squares of its terms is $\frac{64}{3}$. Find r .

- A $-\frac{1}{2}$
- B $\frac{1}{2}$
- C $-\frac{1}{4}$
- D $-\frac{3}{4}$
- E $\frac{1}{3}$
- F $\frac{1}{3}$

16 Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

A 60

B 15

C 160

D 480

E 240

F 120

17 The line L passes through $(4, 0)$ and $(0, 6)$. Find the perpendicular distance from the origin to L .

A 5

B $\frac{24}{13}$

C $\frac{13}{12}$

D $\frac{13}{12}$

E $2\sqrt{13}$

F $\sqrt{13}$

- 18** The graph of $y=f(x)$ has $f'(x) > 0$ for $x < 1$, $f'(1) = 0$, $f'(x) > 0$ for $1 < x < 3$, $f'(x) < 0$ for $x > 3$. Which of the following is true?
- A** f has local maxima at $x = 1$ and $x = 3$
 - B** f has a local minimum at $x = 1$ and a local maximum at $x = 3$
 - C** f has a local maximum at $x = 3$ and a stationary point of inflexion at $x = 1$
 - D** f has a local maximum at $x = 1$ and a local minimum at $x = 3$
 - E** f is increasing for all x

- 19** Find the value of $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{2+\sqrt{3}}$.
- A** 2
- B** $2 - \sqrt{3}$
- C** $\sqrt{3} + 1$
- D** 1
- E** $\sqrt{3}$
- F** $\sqrt{2}$

- 20** A box contains n white balls and 3 black balls. Two balls are drawn without replacement. The probability that both are white is $\frac{5}{12}$. Find n .

- A** 6
- B** 4
- C** 5
- D** 3
- E** 7
- F** 9

END OF TEST